

## **Deliverable #6: Sprint #2**

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## Executive Summary

**To:** Katherine Syvanych, Director - Center for Student Involvement

**Prepared By:** Talent-Link

**Date:** December 4, 2025

Talent-Link builds on our shared vision of a student-facing platform that strengthens peer-to-peer collaboration, highlights the wide range of skills across Kent State, and makes it easier for students to connect, collaborate, and grow together. In the last deliverable, we started work on a prototype, showing you how the system would actually work and function. This deliverable expands the functionality of our earlier prototype and presents our final finding of if this system is feasible and should be developed into a fully-functioning system. Here is an overview of everything that we have completed during this sprint.

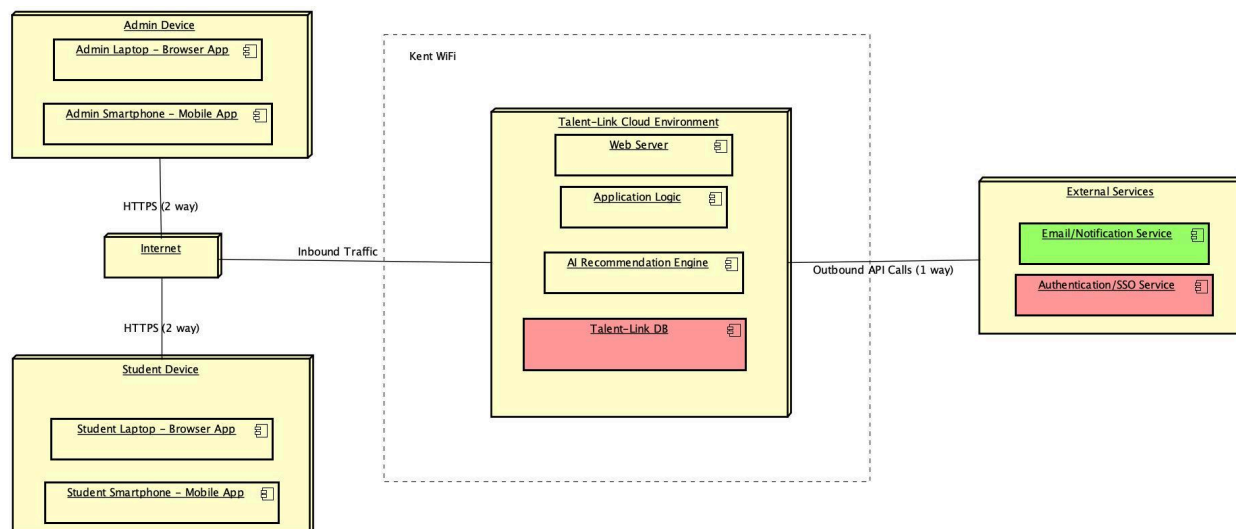
- The updated Sprint Backlog reflects the team's final progress and includes actual completion times for each task in sprint 2. During the last sprint we were able to work on three use cases intended for this sprint (View Sessions, View Analytics, and Suggest Pairs), which reduced the workload in this sprint and allowed us to focus on finalizing the foundation of the system.
- The Network/deployment diagram shows the hardware nodes, as well as the connections between them. This is needed for the more technical aspects of development.
- The Figma Prototype showcases all finalized use cases for the system, including posting, suggesting pairs, matching with peers, viewing sessions, scheduling sessions, managing profiles, sending messages, leaving peer reviews, and moderating student activity for admins. This final prototype now allows CSI to better imagine how students will engage with Talent-Link and decide if it is a good fit for you.
- The accompanying User Manual Videos provide a guided walkthrough of each use case, explaining how users would engage with the system. In our last sprint, we delivered the finalized admin user manual and in this sprint we are delivering the finalized student user manual.
- The Cybersecurity Discussion shows potential security implications and how we suggest to navigate them.
- The Cost/Benefit Analysis gives our final recommendation on if the system should be fully developed and implemented, based on the predicted costs of producing and running the system weighed against the benefits that the system will bring to the university.
- Finally, we will be presenting all of our work on Dec 4, 2025 to showcase our prototype and review the last two sprints. We would like to invite you to attend this presentation so that you can see Talent-Link in action. In case you're unable to attend our presentation, we would like to share that based on the results of our Cost Benefit Analysis, we formally recommend moving forward with the development of Talent-Link, as the projected benefits significantly outweigh the costs and the system demonstrates strong long-term value for CSI and the broader student community.

We invite your feedback on our prototype's final design and hope that you are pleased with the final product.

## Updated Sprint 2 Backlog

| <b>Goal:</b> Prototype "Should Have" use cases and finalize design of overall prototype. |                                                       |                                                |                               |
|------------------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|-------------------------------|
| <b>Name of Task to be Completed</b>                                                      | <b>Team Member(s) Assigned Primary Responsibility</b> | <b>Estimated Effort for the Task (XXS-XXL)</b> | <b>Actual Completion Time</b> |
| Send Message Prototype                                                                   | Grace, Yusur                                          | M                                              | 1 hour                        |
| Leave Peer Review Prototype                                                              | Yusur                                                 | M                                              | 1 hour                        |
| Video User Manual (student)                                                              | Jordan                                                | L                                              | 10 minutes                    |
| Network/Deployment Diagram                                                               | Lucy                                                  | L                                              | 1.5 hours                     |
| Cybersecurity Discussion                                                                 | Grace                                                 | M                                              | 1 hour                        |
| Cost/Benefit Analysis                                                                    | Akaiya                                                | L                                              | 1 hour                        |
| Presentation Slides                                                                      | All                                                   | XL                                             | 30 minutes                    |
| D6 Executive Summary                                                                     | Jordan                                                | S                                              | 30 minutes                    |
| D6 Appendix of Team Member Contributions                                                 | All                                                   | XXS                                            | 10 minutes                    |
| Update Sprint Backlog with actual completion times                                       | All                                                   | XS                                             | 10 minutes                    |
| Compile and finalize deliverable for submission                                          | All                                                   | M                                              | 20 minutes                    |
| Report User Prototype                                                                    | Yusur & Jordan                                        | S                                              | 15 minutes                    |

## Network / Deployment Diagram



This network deployment diagram shows how the Talent-Link system interacts across the different devices and services that support it. Both students and administrators can access Talent-Link by their laptops or mobile apps, and their requests travel safely through the internet using HTTPS. These requests then pass through the system's firewall, which protects the internal cloud environment where the main parts of the Talent-Link System run. Inside the secure zone, the web server handles incoming traffic, the application logic manages system functions, and the AI recommendation engine helps match users based on their skills and interests. All profile, skill, post, and connection information is stored in the Talent-Link database, which is kept within the protected area since it contains sensitive data. The system also communicates with external services outside of the firewall, such as the authentication/SSO service for login via FlashLine, as well as the email/notification service that sends alerts to users. Together, these nodes and connections support a secure and efficient environment that allows users and administrators to use Talent-Link reliably from any device.

## Final Prototype



### Link to Prototype:

<https://www.figma.com/proto/F1X6EqtoYNcKw5iJeA13Oy/Talent-Link-Prototype?page-id=0%3A1&node-id=3-30&p=f&viewport=76%2C437%2C0.04&t=noqqMhbALB8RbFjp-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=3%3A30&show-prototype-sidebar=1>

### New Features Completed in this Sprint:

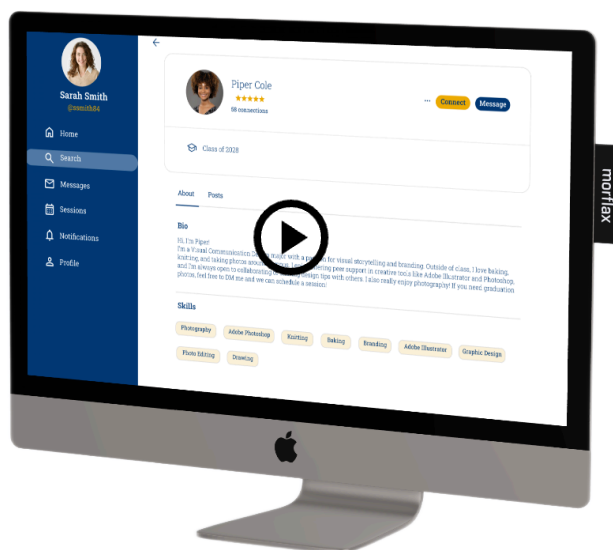
- Leave Peer Review in “Sessions” page
- Send Message in “Messages” page, including starting a new chat
- Notifications page
- The option to report peers from different pages
- The option to view your connections in your profile without having to navigate to “manage profile”

## Video User Manual

### Link to User Manual Video:

[https://video.kent.edu/media/Talent%20Link%20User%20Manual/1\\_bayspofe](https://video.kent.edu/media/Talent%20Link%20User%20Manual/1_bayspofe)

\*\* For better viewing, hit the toggle view button in the top right corner of the video and then the picture in picture button below the toggle view button and click the single view to only see the screen recording.



## Cybersecurity Discussion

The Talent-Link system faces several cybersecurity risks based on how information flows through the nose in the deployment diagram. The most significant threat is a data breach involving the Talent-Link database, which serves as a central repository for sensitive student information such as names, skills, bios, and connection history. If attackers were to gain access to this database, they could expose or manipulate personal data, causing major privacy, reputational, and compliance issues for Kent. Authentication vulnerabilities pose another risk. If weaknesses exist in the Authentication/SSO Service or the login process, attackers could bypass authentication entirely and impersonate students or administrators, gaining unauthorized access to Talent-Link's internal features and data. However, since the login is through FlashLine, we have full confidence in the system that Kent State already has in place.

Traffic between student or administrative devices and the Talent-Link Cloud Environment also introduces exposure. Since this communication travels across the Internet and potentially the public WiFi, it is vulnerable to man-in-the-middle (MITM) attacks if the network is not secured properly. An attacker could potentially intercept conversations, steal credentials, or alter data before it reaches the web server. The Web Server and Application Logic themselves are also targets for denial-of-service (DoS) attacks, where an attacker can overload the system with traffic, preventing real users from accessing Talent-Link. If either of these nodes were to go down, students and administrators would be unable to log in or do anything on the platform.

To reduce these risks, strong security measures must be implemented at each layer shown in the diagram. Encrypting all communication using HTTPS helps protect against MITM threats, while strict firewall rules can limit unauthorized access into the cloud environment. As stated above, the authentication process is through FlashLine, and users/administrators can only access Talent-Link through a button in FlashLine, so they must be logged in first in order to use our platform. Rate limiting and intrusion detection systems on the Web Server can help strengthen against DoS attempts. Together, these safeguards strengthen the security of each node and connection in the system and improve the overall integrity and resilience of Talent-Link.

## Cost Benefit Analysis

| Year         | Requirements Determination | Analysis & Design | Development        | Installation      | Operation & Maintenance | Yearly Costs       | PV of Yearly Costs | Yearly Benefits   | PV of Yearly Benefits | NPV (PV Benefits - PV Costs) | Cumulative Total |
|--------------|----------------------------|-------------------|--------------------|-------------------|-------------------------|--------------------|--------------------|-------------------|-----------------------|------------------------------|------------------|
| 1            | (8,250)                    | (5,000)           | (16,000)           | (2,700)           |                         | (31,950)           | (31,950)           | -                 | -                     | (31,950)                     | (31,950)         |
| 2            |                            |                   |                    |                   | (7,200)                 | (7,200)            | (6,486)            | 49,200            | 44,324                | 37,838                       | 5,888            |
| 3            |                            |                   |                    |                   | (7,200)                 | (7,200)            | (5,844)            | 49,200            | 39,932                | 34,088                       | 39,976           |
| 4            |                            |                   |                    |                   | (7,200)                 | (7,200)            | (5,265)            | 49,200            | 35,975                | 30,710                       | 70,686           |
| 5            |                            |                   |                    |                   | (7,200)                 | (7,200)            | (4,743)            | 49,200            | 32,410                | 27,667                       | 98,353           |
| <b>Total</b> | <b>\$ (8,250)</b>          | <b>\$ (5,000)</b> | <b>\$ (16,000)</b> | <b>\$ (2,700)</b> | <b>\$ (28,800)</b>      | <b>\$ (60,750)</b> | <b>\$ (54,288)</b> | <b>\$ 196,800</b> | <b>\$ 152,640</b>     | <b>\$ 98,353</b>             |                  |

We decided on a Discount Rate (Required Rate of Return) of 11% based on the current Prime Rate of 7.00% (fedfundrate.org, retrieved November 25, 2025) plus a 4% risk premium. Talent-Link involves moderate risk because it introduces new student-facing functionality and leverages AI matching, but operates in a stable university environment. This rate reflects both competitive risk and the opportunity cost of allocating resources to a new system.

- We estimated our requirements from the cost of labor and the tools needed to host and operate the system. An internal KSU IT team would build the system, and Kent State employees working in application development earns a yearly salary of approximately \$72,400. This calculates to around \$35 an hour, and for a total development period of 30 hours with a team of five would create a labor cost of \$5,250 ([Buckeye Institute](#)). Costs for web-hosting the system would be approximately \$250 per month, which totals to \$3,000 a year ([Atomic Interactive](#)), which estimated our total requirements to cost \$8,250.
- The estimated cost for the analysis and design of Talent-Link would be approximately \$5,000 since it has around ten pages across both interfaces, and the interfaces already have a basic design ([Forefront Web](#)).
- Development would be estimated at \$13,000 based on the size and complexity of the system, and with an ongoing SEO upkeep plan of approximately \$250 per month, our total development cost would be \$16,000 for the first year ([The Real Social Company](#)).
- Installation costs would be approximately \$1,500 upfront, and including the monthly maintenance cost of \$100 a month, the total cost would be \$2,700 ([Lead Pages](#)).

Our annual costs after the first year would be \$7,200 based on the different monthly upkeep costs from web-hosting, maintenance, and SEO. Our annual benefits after installation would conservatively round out to \$49,200 based on the estimated that if the platform helps retain four students per year who would otherwise leave due to lack of campus connection, Kent State would earn \$49,217.60 yearly in tuition revenue (4 students × \$12,304.40 tuition per student).

Talent-Link will achieve break-even in Year 2, with cumulative NPV becoming positive at a profit of \$5,888. At Year 5, Talent-Link will have earned approximately \$100,000 in profit. This quick payback period indicates low financial risk and demonstrates that retention benefits quickly outweigh the initial development and ongoing operational costs.

Intangible costs to consider include the potential learning curve for both students and staff when the platform is first launched, the temporary increase in help-desk tickets that would come with the early-release and still foreign software, and the need for consistent content moderation by administrators.



However, the intangible benefits outweigh the cost in that Talent-Link will help to increase student satisfaction and community engagement, expand access to skill-building and networking opportunities, and heighten visibility for stakeholders and administrators to understand emerging student talents and currently popular interests.

Our formal recommendation would be to proceed with the system as planned, as our benefits are more than evidence enough to establish grounds for a profitable and beneficial platform. The low payback period to achieve profits, along with the revenue gained from student retention, establishes a low-risk and high-reward system that should continue to be developed and deployed.

### Appendix of Team Member Contributions

| Team Member     | Percentage of Overall Work | Contributions                                                                                                                                                                                                              |
|-----------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grace Karpinski | 20%                        | <ul style="list-style-type: none"> <li>- Completed the Cybersecurity Discussion</li> <li>- Contributed to Send a Message prototype</li> </ul>                                                                              |
| Yusur Alrawi    | 20%                        | <ul style="list-style-type: none"> <li>- Completed leave peer review prototype</li> <li>- Completed Send Message prototype</li> <li>- Completed notifications page</li> <li>- Finalized slides for presentation</li> </ul> |
| Lucy Hennessy   | 20%                        | <ul style="list-style-type: none"> <li>- Completed the network diagram</li> <li>- Helped finalize deliverable</li> <li>- Finalized team member contributions</li> </ul>                                                    |
| Akaiya Abdullah | 20%                        | <ul style="list-style-type: none"> <li>- Completed the Cost-Benefit Analysis</li> <li>- Completed intangible benefits and costs discussion</li> </ul>                                                                      |
| Jordan Jenkins  | 20%                        | <ul style="list-style-type: none"> <li>- Completed Video User Manual</li> <li>- Completed Executive Summary</li> <li>- Helped with report user prototype and finalizing the deliverable</li> </ul>                         |

1. **If your semester-long project experience was a candy bar, what kind would it be and why? (This is a relatively fun question frequently used in real-life Scrum retrospectives.)**
  - a. Kit Kat. The project naturally broke into clean and manageable pieces. Each deliverable helped us set the layout for the next one, and eventually we got to Talent-Link where we put all the pieces together.
2. **Scrum emphasizes collective – as opposed to individual team member – accountability. How well does your team think this principle works in an academic (as opposed to industry) context, and why? How successful was your own team in living up to this principle?**
  - a. Our team thinks scrum works very well in an academic environment, specifically with this project. Since the bulk of the work was split into two different deliverables, we were able to split up the work in the sprint backlog so we knew what was expected before we began Deliverable 5. We did a decent job following our sprint backlog. However, we noticed after the completion of Deliverable 5, we finished almost all of the prototype, therefore lessening the work we had to complete for Deliverable 6. We still think the sprint backlogs and Scrum as a whole worked very well, as we were able to divide up the work based on what was expected to take the most amount of effort. This made it easier for us to work together, since one person was not stuck with all of the hardest, most time-consuming tasks.

3. **Were there any extenuating circumstances that led to an uneven distribution of the workload in this final sprint? Was your team as a whole satisfied with the overall workload carried by each team member, accounting for occasional variability across individual deliverables or sprints? (Explain.)**
  - a. The only circumstances would be life/jobs outside of class, but as a whole, we all got our work done in a timely manner, and we did good quality work. The whole team worked together before each deliverable to divide up the work to make sure everything was evenly split up. We had constant communication so if one person was behind we would help each other with picking up in different areas as we needed.
4. **Did you use any AI tools or other resources to generate or proofread any part of your final draft? If so, explain here (tell me what SW or resources you used, & precisely what they were used for).**
  - a. No AI was used in this deliverable.